

HPDP PROJECT 2

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1.0 Introduction

- huge increase in user-generated content such as online reviews, comments, and social media posts.
- provide insights into public opinions, experiences, and preferences regarding products, services, and tourist destinations.
- Malaysia, as a major tourism destination in Southeast Asia, receives thousands of visitor reviews through platforms such as Google Maps.
- These reviews contain important information about:
 - Visitor satisfaction
 - Service quality
 - Overall travel experiences



2.1 Objectives

- Collect English-language Google Maps reviews from Malaysian tourist attractions via automated web scraping
- Label and preprocess review data using a pre-trained transformer-based model
- Train and evaluate deep learning and traditional machine learning models for 3-class sentiment classification
- Design and implement a real-time streaming pipeline for automated sentiment classification
- Develop a Kibana dashboard for real-time visualisation of sentiment results



2.2 Scope

- Dataset: 3,271 reviews from 13 Malaysian tourist attractions
 - Batu Caves, Petronas Twin Towers, Pulau Sipadan, Mossy Forest, Niah National Park, Genting SkyWorld, Taman Negara, Kek Lok Si Temple, and others
- Three sentiment classes: Positive, Neutral, Negative (English-language only)
- Primary model: CNN for real-time inference
- Streaming via Apache Kafka; storage via Elasticsearch; visualisation via Kibana
- Fully containerised using Docker Compose

3.1 Data Collection

- Source: Google Maps reviews of 13 Malaysian tourist attractions
- No public API → custom scraper built using Selenium with headless Chrome
- Dynamic scrolling to load and extract reviews
- Fault-tolerance: exponential backoff retries + resumable state via progress.json
- Raw dataset: 3,271 reviews — columns: place_url, review_text
- Labels assigned using Hugging Face cardiffnlp/twitter-roberta-base-sentiment-latest
- RoBERTa-based transformer model pre-trained for sentiment classification

Sentiment	Count	Share
Positive	2533	78%
Neutral	488	14%
Negative	261	8%



3.2 Data Preprocessing

- Steps applied sequentially to each review:
 1. Fix text encoding artefacts using ftfy
 2. Convert to lowercase
 3. Remove emojis, URLs, special characters, and numbers
 4. Tokenise using NLTK word_tokenize
 5. Remove English stopwords (NLTK corpus)
 6. Lemmatise tokens using WordNetLemmatizer
- Output: cleaned_data.csv with columns clean_review and label
- 29 empty reviews discarded → final dataset: 3,242 reviews

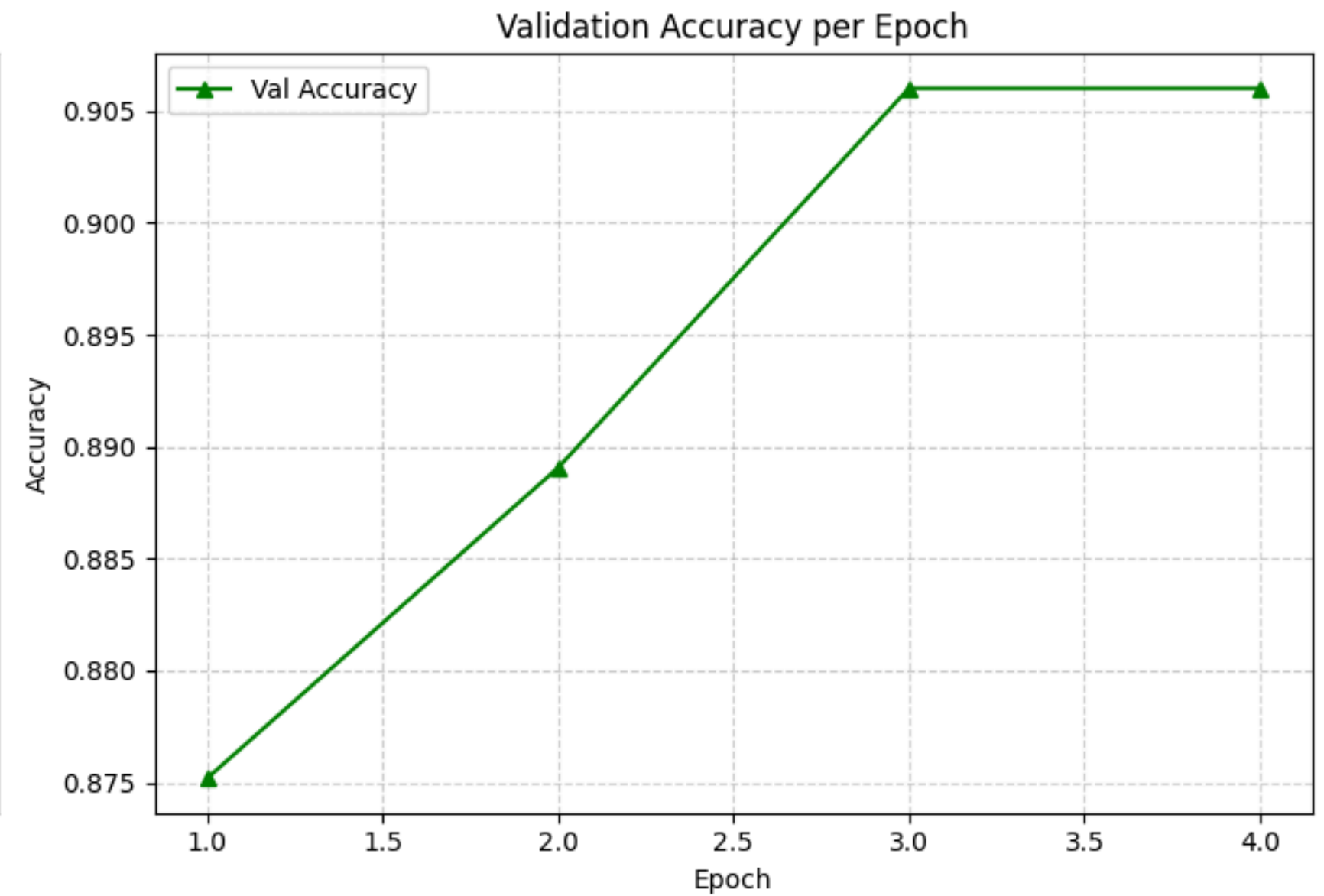
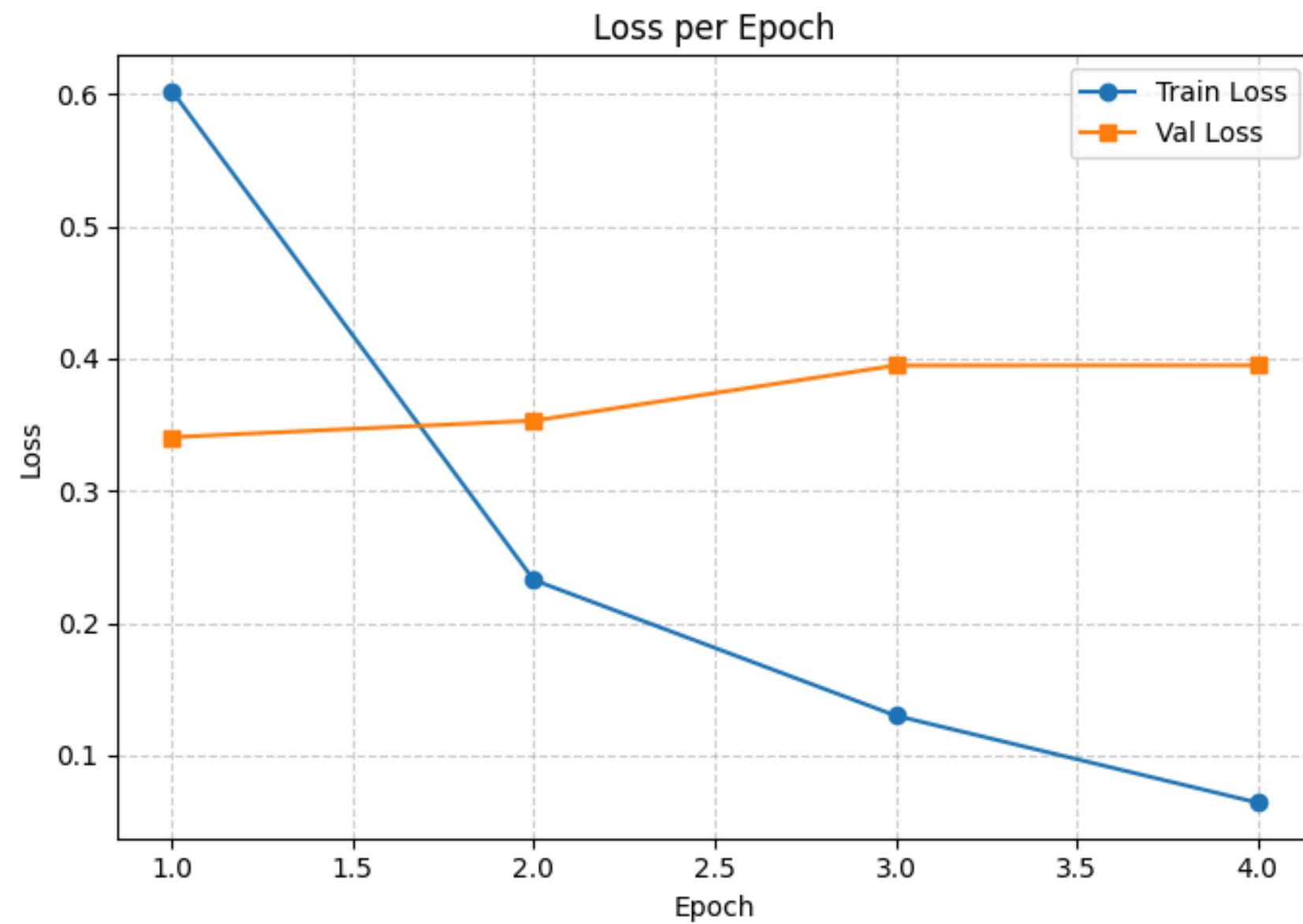
4.1 BERT

4.1.1 Model Choice

- bidirectional attention mechanism
- capturing the linguistic meaning and context of review text more effectively
- Contains 12 transformer layers.
- Has approximately 110 million parameters.
- Converts all input text to lowercase.
- Removes accent marks during preprocessing.
- Suitable for informal Google Maps reviews.
- Provides better generalisation on textual data.
- Requires lower memory usage compared to larger BERT models.

4.1 BERT

4.1.2 Training Process





Group: Shopping

4.1.3 Evaluation

Classification Report of BERT sentiment model

TEST SET METRICS (macro avg)

Accuracy : 0.8752

Precision : 0.7286

Recall : 0.7532

F1-Score : 0.7374

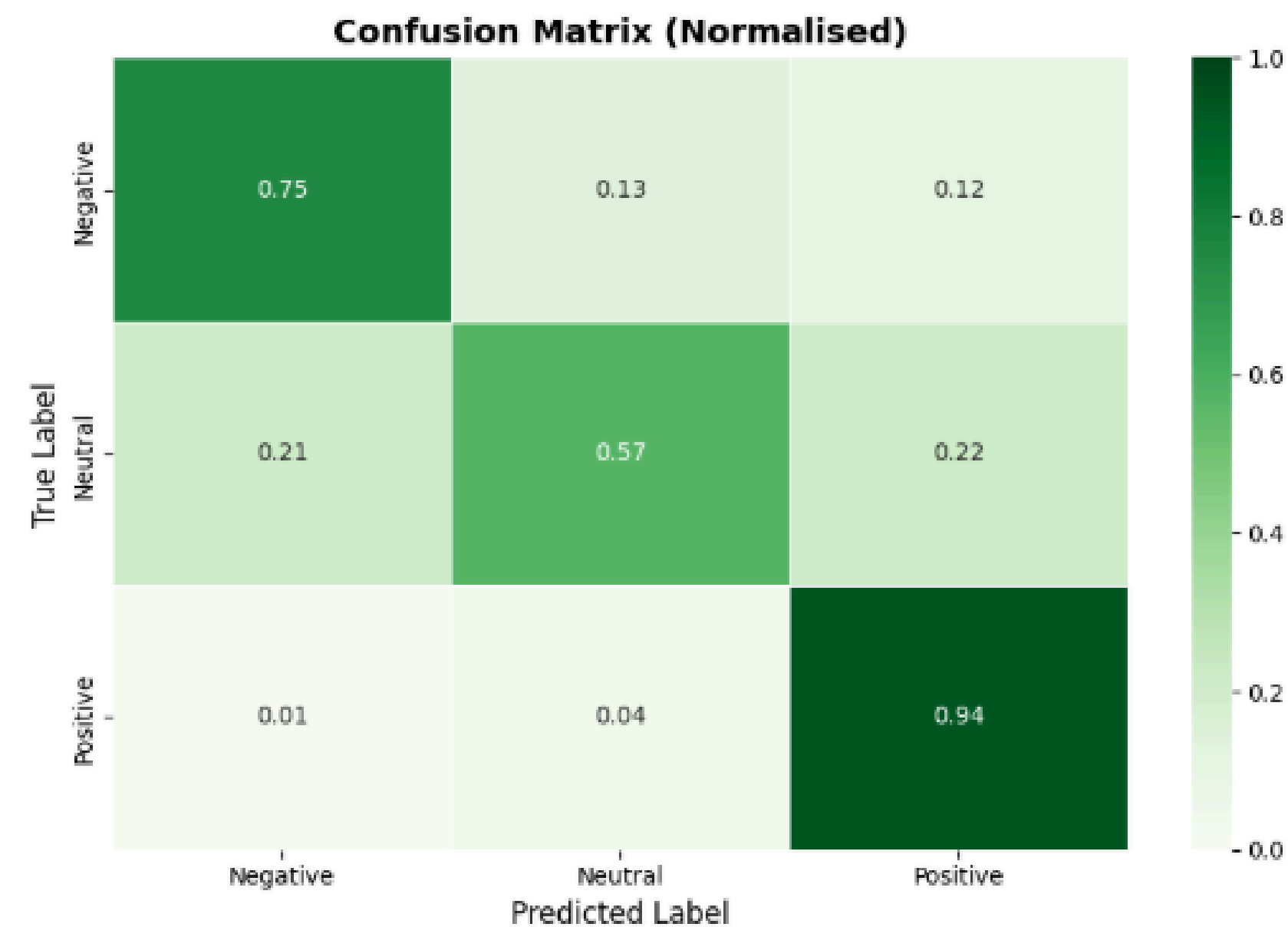
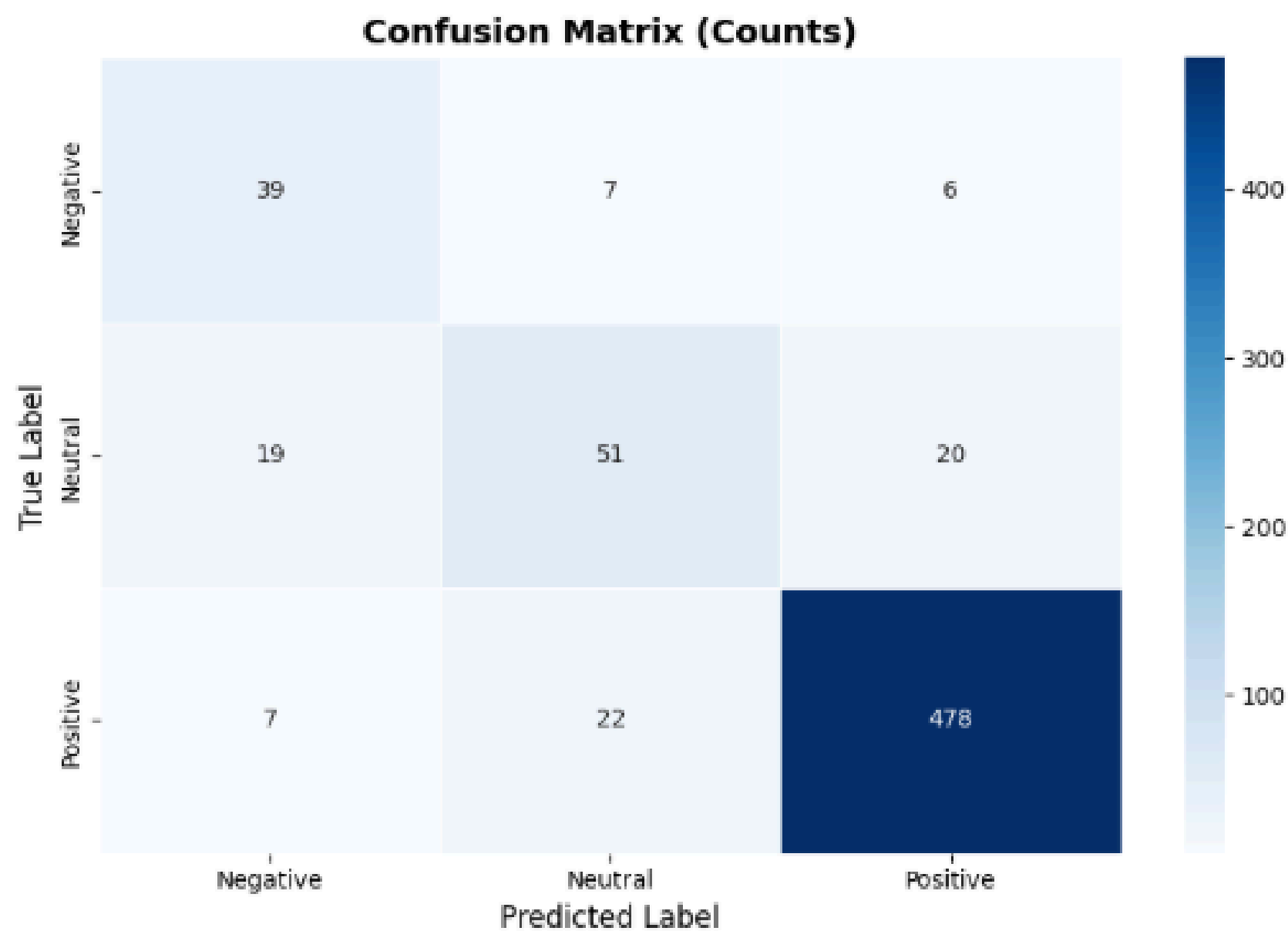
Detailed Classification Report:

	precision	recall	f1-score	support
Negative	0.60	0.75	0.67	52
Neutral	0.64	0.57	0.60	90
Positive	0.95	0.94	0.95	507
accuracy			0.88	649
macro avg	0.73	0.75	0.74	649
weighted avg	0.88	0.88	0.88	649

4.1.3 Evaluation

Confusion matrix of BERT sentiment model

BERT Sentiment Classification | Test Accuracy: 0.8752



4.2 CNN

4.2.1 Model Choice

- **effective for text classification tasks**

CNN's convolutional filters are able to capture local n-gram features from the sequences of token embeddings -> helps the model to capture the sentiment bearing phrases irrespective of their position in the text.

- **has much fewer parameters**

Practical for deployment in a streaming pipeline.

- **can use significantly less memory and computation**

The inference process should be performed continuously with low latency.

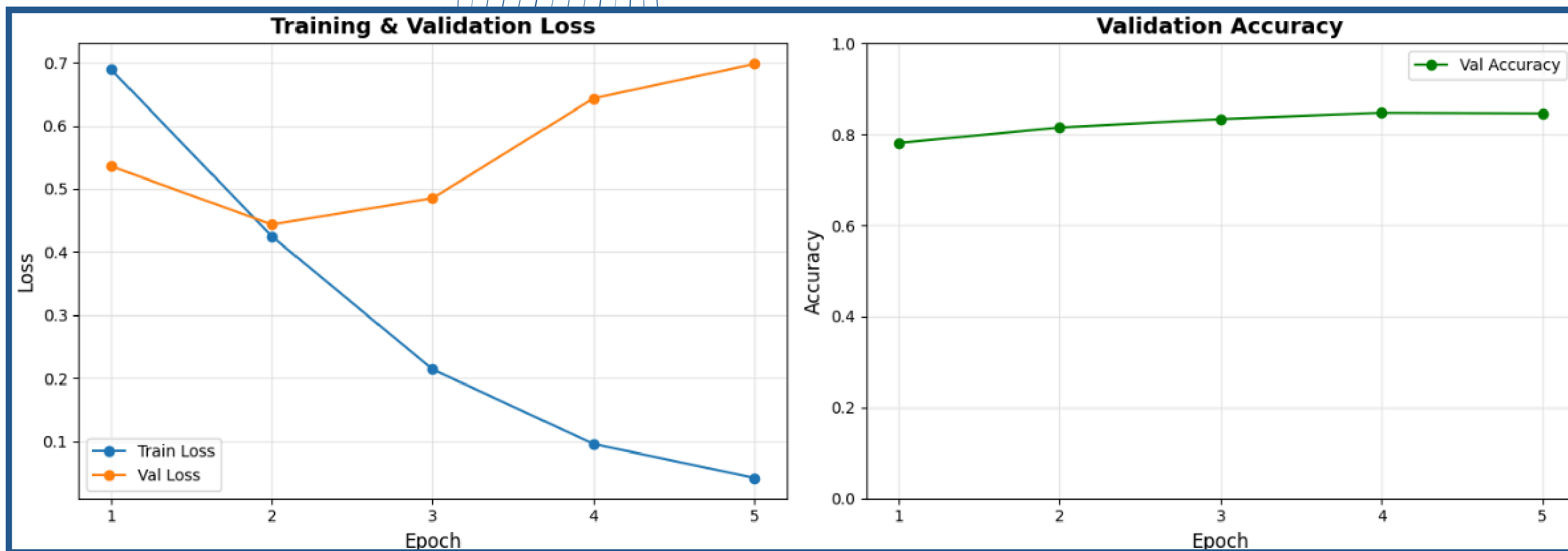
- **operate efficiently over CPU for real-time inference**

The used architecture prevent overfitting while the embedding layer was trained from scratch.



4.2 CNN

4.2.2 Training Process



4.2 CNN

4.2.3 Evaluation

Classification Report of CNN sentiment model

The overall test accuracy of the CNN is 81.51%, while the accuracy of the BERT model is 87.51%. The higher the F1 score, the better. However, due to the large class imbalance in this dataset, the macro-averaged F1 score is 0.5261, indicating that it is hard to classify the minority classes.

Loaded best model from 'best_cnn_sentiment.keras'

TEST SET METRICS (macro avg)

Accuracy : 0.8151
Precision : 0.6566
Recall : 0.5289
F1-Score : 0.5261

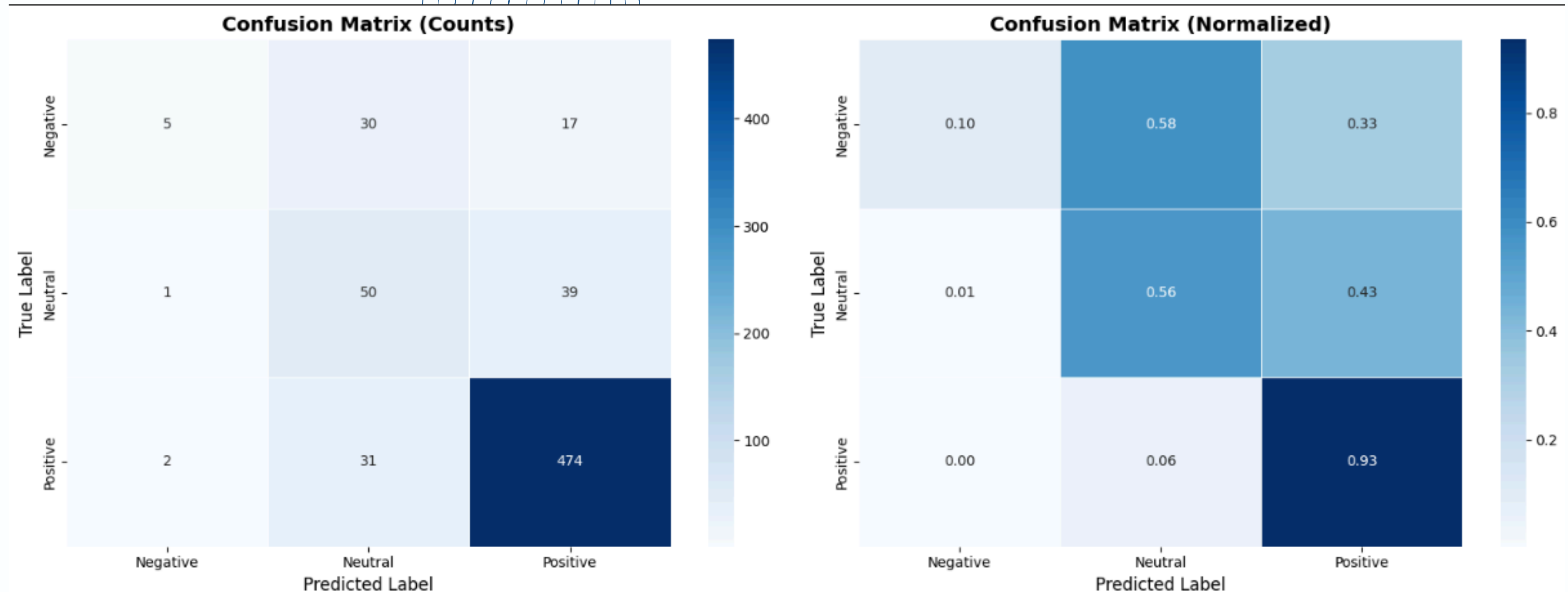
Detailed classification report:

	precision	recall	f1-score	support
Negative	0.62	0.10	0.17	52
Neutral	0.45	0.56	0.50	90
Positive	0.89	0.93	0.91	507
accuracy			0.82	649
macro avg	0.66	0.53	0.53	649
weighted avg	0.81	0.82	0.80	649

4.2 CNN

4.2.3 Evaluation

Confusion matrix of CNN sentiment model

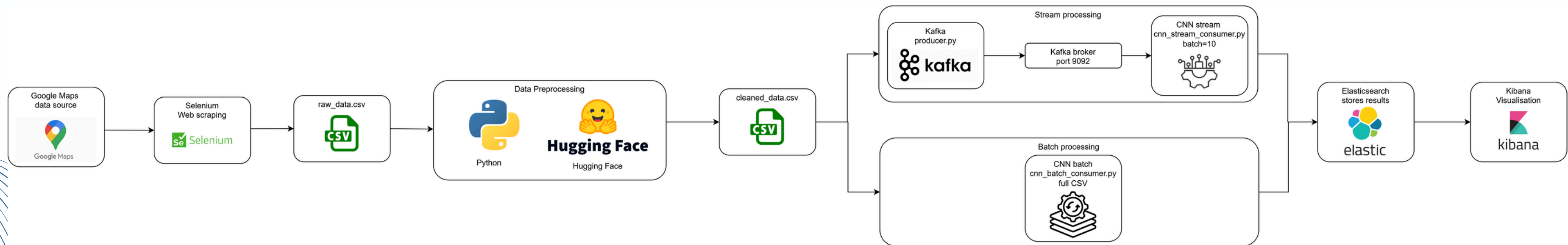


4.3 Model Comparison

Features Comparison	BERT	CNN
Accuracy	87.52%	81.51%
Macro Precision	0.7286	0.6566
Macro Recall	0.7532	0.5289
Macro F1-Score	0.7374	0.5261
Parameters	~110 million	1,028,483
Inference Speed	Slow (transformer attention)	Fast (single Conv1D pass)
Memory Usage	High	Low
Deployed in Pipeline	No	Yes

ransform

5.0 Apache System Architecture



6.0 Analysis & Key Findings

6.1 Key Findings

Sentiment Diffusion

Model Performance

Batch vs Streaming
Throughput

Most Frequently
Discussed Topic

Resource Usage

6.0 Analysis & Key Findings

6.2 Visualization



6.0 Analysis & Key Findings

6.3 Insights

Tourist sentiment is predominantly positive.

Class imbalance is the primary limitation of the CNN model.

Batch processing suits fixed-volume retrospective analysis

Streaming processing enables real-time monitoring with manageable overhead

Duplicate records limit the reliability of Kibana aggregations

7.0 Optimization and Comparison

- **knowledge distribution and model quantisation**
- **Kafka Partitioning and Consumer Group Scaling**
- **Elasticsearch Bulk Indexing and Asynchronous Processing**

- **Lower memory consumption, faster inference, better classification performance.**
- **Higher throughput through parallel processing and improved scalability.**
- **Reduced network overhead and improved indexing throughput.**

8.0 Conclusion & Future Work

This project built a real-time sentiment analysis pipeline using Apache Kafka, CNN, Elasticsearch, and Kibana on 3,242 Google Maps reviews from 13 Malaysian tourist attractions. CNN (81.51% accuracy) was chosen over BERT (87.52%) for deployment due to its lightweight architecture, achieving ~115 rec/sec streaming and ~126 rec/sec batch throughput. Results confirmed predominantly positive tourist sentiment across Malaysia. Future work includes replacing CNN with DistilBERT, applying SMOTE for class imbalance, and expanding to multilingual data and more platforms.